

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

IPv6 adoption strategies

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Drivers for IPv6 adoption



- Government and Policy Mandates
- Public IPv4 exhaustion
- Private IPv4 exhaustion

Coming in 2024

Public IPv4 pricing

Why?

- IPv4 addresses are an increasingly scarce resource

What does this mean for you?

- AWS will start charging for public IPv4 address from Feb 1, 2024
- Charge of \$0.005 per IP per hour (\$43.8 per IP per year) for all public IPv4 addresses, whether attached to a service or not
- Be frugal with your use of public IPv4 addresses
- Another business justification to accelerate your adoption of IPv6
- Leverage Bring Your Own IP (BYOIP) for your use-case
- Monitor, analyze, and audit public IPv4 usage using Public IP Insights across AWS Organizations

IPv6 support for AWS Services



Amazon EC2



Amazon S3



Amazon VPC



Amazon CloudFront



Elastic Load Balancing (ELB)



Amazon Lightsail



IAM



AWS Transit Gateway



AWS IoT



Amazon ECS



AWS Shield



Amazon Route 53



AWS WAF



AWS Site-to-Site VPN



Amazon EKS



Amazon WorkSpaces



AWS Direct Connect

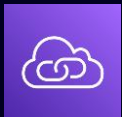


AWS Fargate

New in 2022



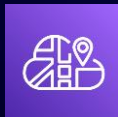
AWS Global Accelerator



AWS PrivateLink



Amazon Athena



AWS Cloud Map



AWS App Mesh



Amazon RDS



Amazon Aurora



AWS DMS



AWS Lambda



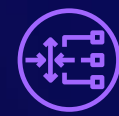
AWS Secrets Manager



Amazon ElastiCache



Amazon EBS



Gateway Load Balancer



What's new in 2023



Amazon Athena

- Supports IPv6 through AWS PrivateLink for inbound connections (May 2023)



AWS
Global Accelerator

- Supports dual-stack NLB endpoints (November 2023) and IPv6 for EC2 endpoints (August 2023)
- Supports TCP termination at the edge for IPv6 traffic (May 2023)



AWS
Network Firewall

- Supports IPv6-only subnets (April 2023) and dual-stack subnets (January 2023)



Amazon VPC

- VPCs and subnets now support more sizes for IPv6 CIDRs (November 2023)
- Supports primary IPv6 address on an elastic network interface (August 2023)
- Amazon VPC IP Address Manager (IPAM) supports AWS-provided contiguous IPv6 CIDR blocks (January 2023)
- Introducing Free Tier and AWS Organization-wide Public IP Insights within Amazon VPC IP Address Manager (IPAM) (November 2023)



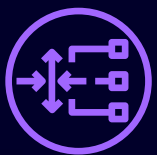
Amazon
Route 53 Resolver

- Supports IPv6-only and dual-stack subnets for inbound and outbound resolver endpoints (March 2023)

What's new in 2023



IPv6 service
support dashboard



Gateway Load
Balancer

- Deploy Gateway Load Balancer Endpoint between virtual private gateway and subnets in Amazon VPC to route IPv4 and IPv6 traffic (August 2023)



AWS IoT Core

- Announces a new inline Rule function that enables you to extract the source IP of an IT client (September 2023)



Elastic Load
Balancing (ELB)

- Application Load Balancer and Network Load Balancer support registering instances addressed by IPv6 as targets (October 2023)



AWS Lambda

- Supports IPv6 for outbound connections in VPC (October 2023)



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Amazon OpenSearch
Service

- Supports IPv6 for new and existing domains (October 2023)



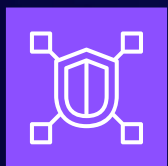
AWS App Runner

- Supports IPv6 for public inbound traffic (November 2023)



Amazon Elastic Kubernetes
Service (Amazon EKS)

- Amazon VPC CNI supports IPv6 Egress for Pods in IPv4 enabled Kubernetes Clusters (June 2023)

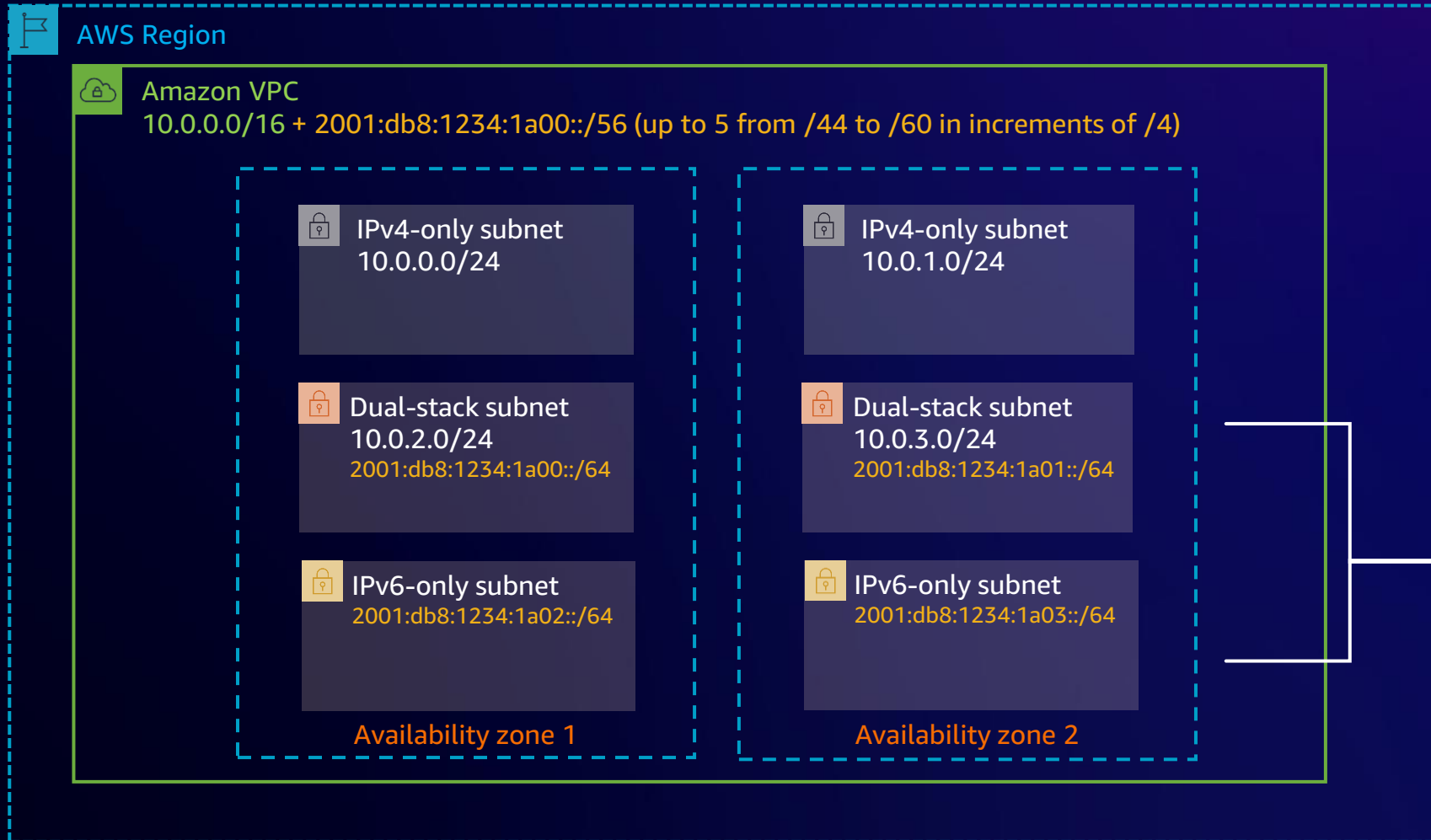


Amazon VPC Lattice

- Supports IPv6-only and dual-stack endpoints for inbound and outbound resolver endpoints (March 2023)

Dual-stack Amazon VPC – Subnet types

IPV4-ONLY, DUAL-STACK, AND IPV6-ONLY SUBNETS



Choose subnet CIDR from /44 to /64 in increments of /4

Choose an IPv6 addressing scheme

- Design your IPv6 addressing strategy using **Amazon VPC IP Address Manager (IPAM)**
- Choose between **Bring your own IPv6 (BYOIPv6)** and **AWS-provided IPv6 CIDR block**
- **Contiguous** IPv6 blocks with BYOIPv6 and AWS-provided IPv6 CIDR block

Public scope

Amazon-provided
CIDRs

Top-level pool

2001:db8:0000::/48

Regional pool

us-west-1

2001:db8:0000::/52

Regional pool

eu-west-2

2001:db8:0100::/52

Regional pool

ap-southeast-2

2001:db8:0200::/52

Dev pool
/54

Test pool
/54

Prod pool
/54

Dev pool
/54

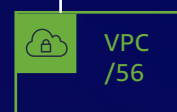
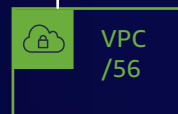
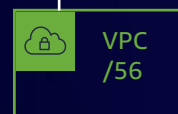
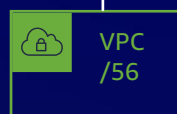
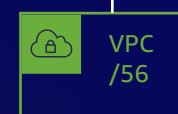
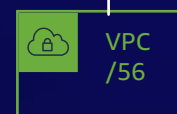
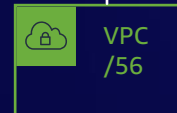
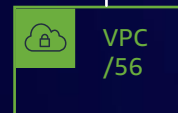
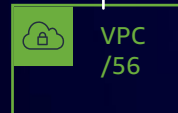
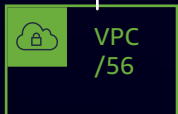
Test pool
/54

Prod pool
/54

Dev pool
/54

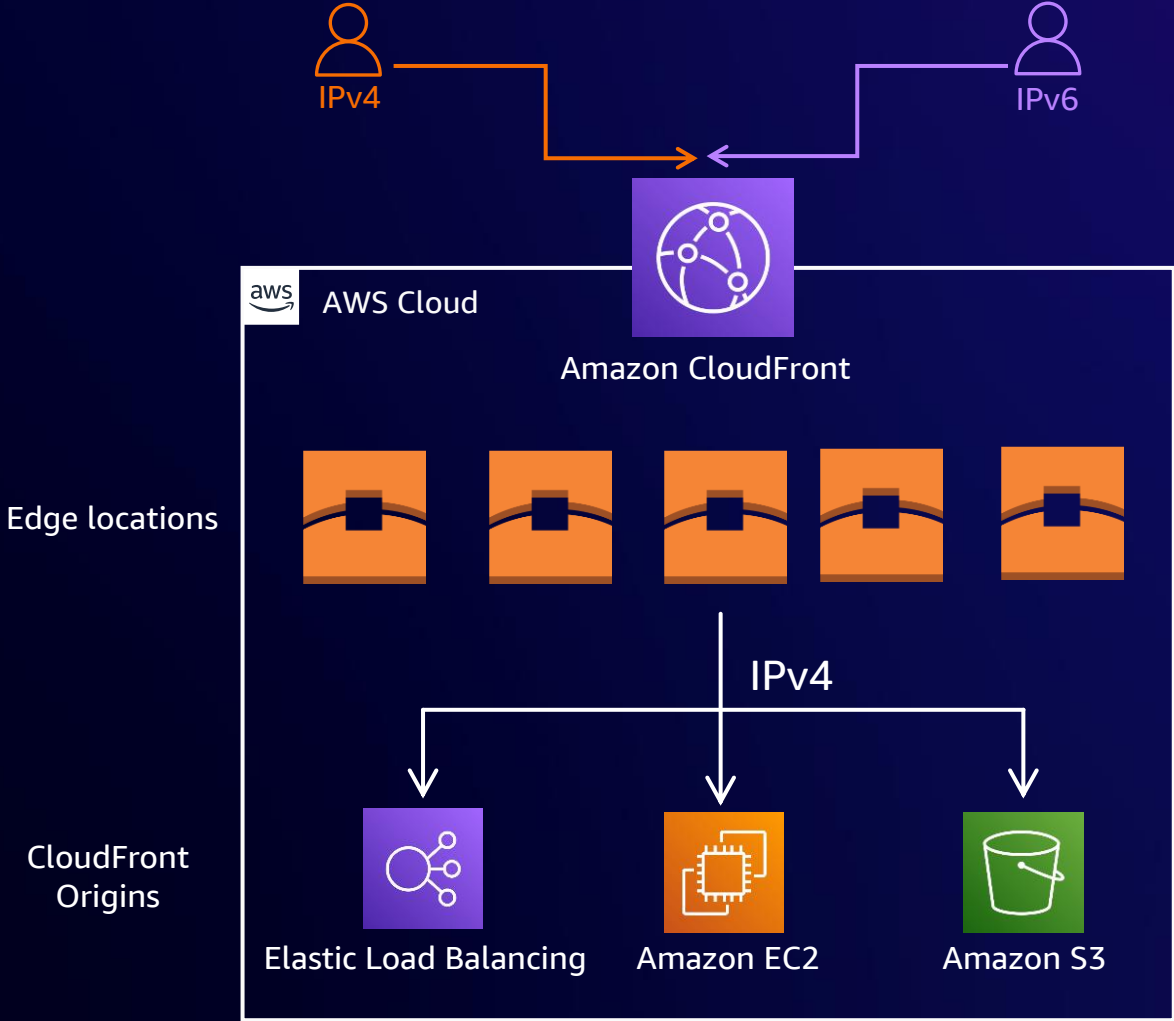
Test pool
/54

Prod pool
/54



Strategy 1: Outside in

AMAZON CLOUDFRONT



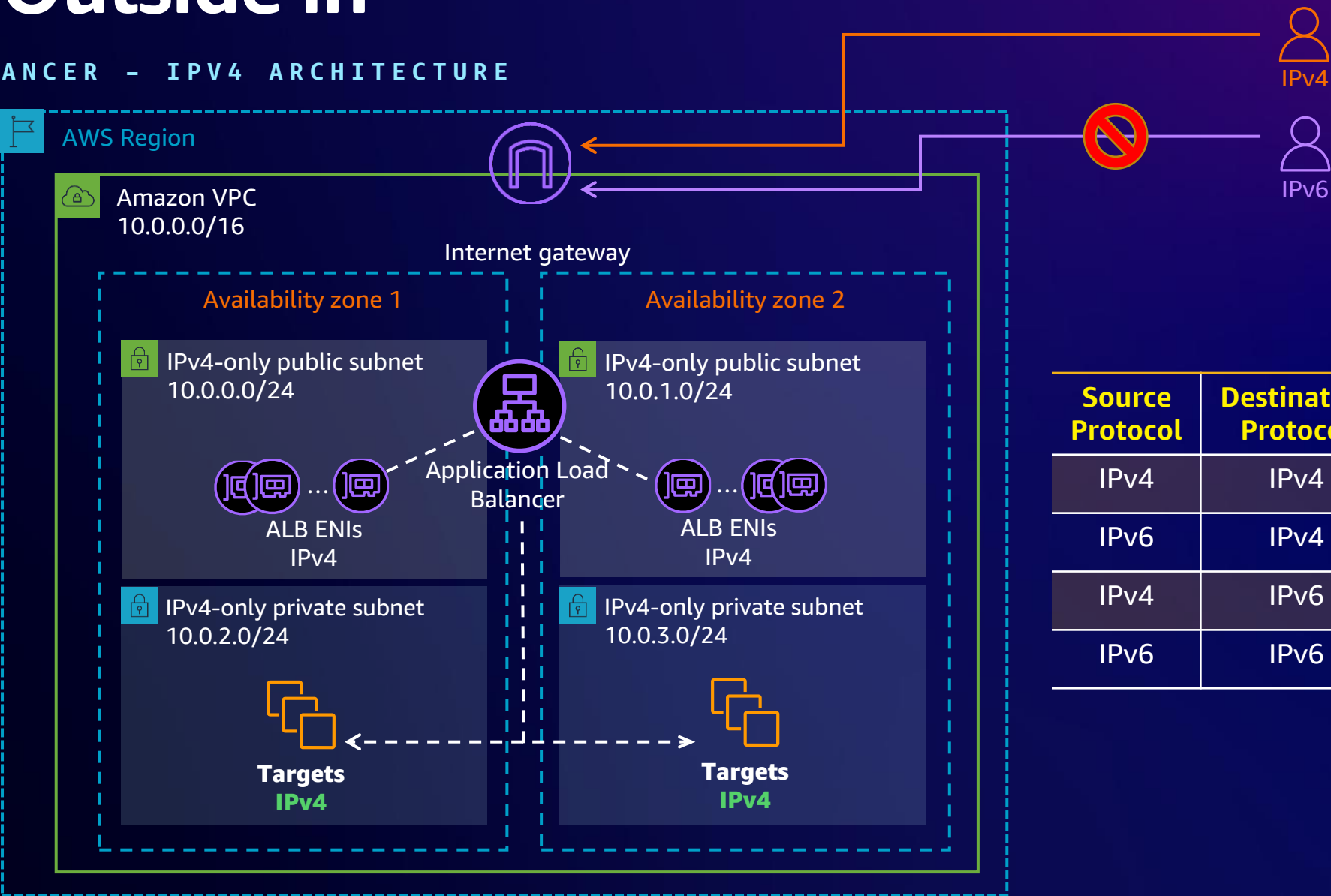
192.0.2.10
192.0.2.45
2001:db8:0042::ff00
2001:db8:ff00:0042

Source Protocol	Destination Protocol	
IPv4	IPv4	✓
IPv6	IPv4	✓
IPv6	IPv6	✗
IPv4	IPv6	✗



Strategy 1: Outside in

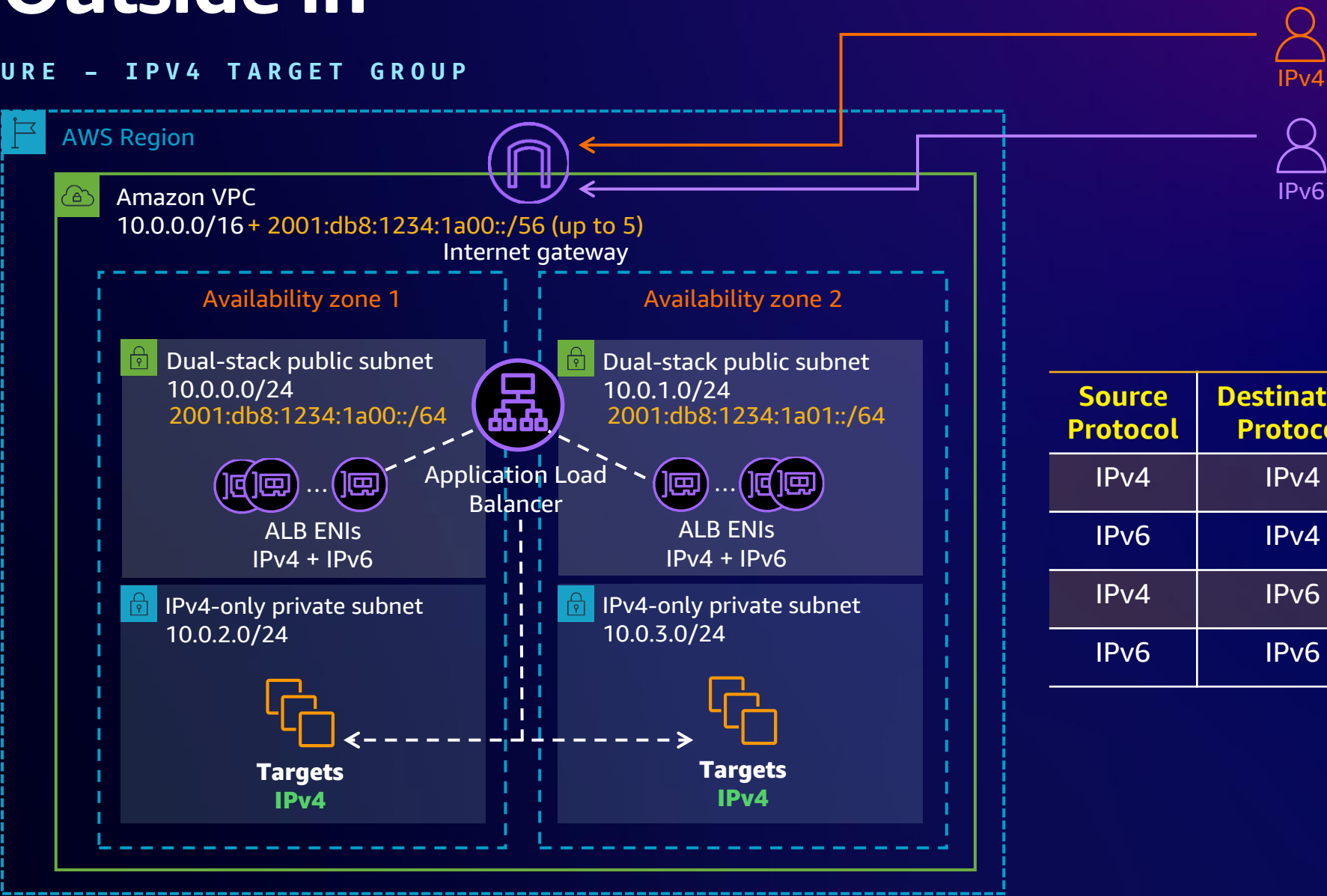
APPLICATION LOAD BALANCER - IPV4 ARCHITECTURE



Source Protocol	Destination Protocol	
IPv4	IPv4	✓
IPv6	IPv4	✗
IPv4	IPv6	✗
IPv6	IPv6	✗

Strategy 1: Outside in

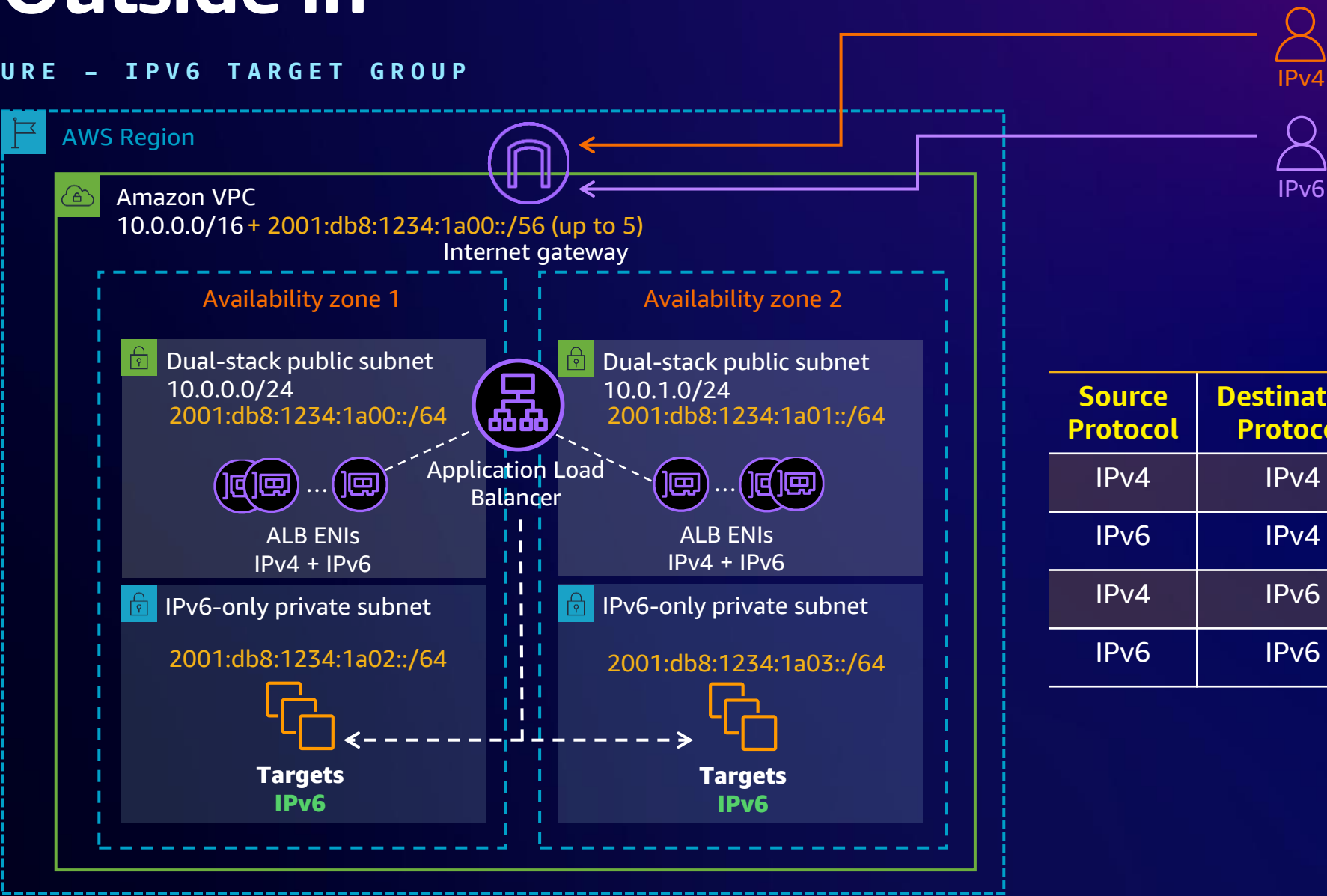
DUAL-STACK ARCHITECTURE - IPV4 TARGET GROUP



Source Protocol	Destination Protocol	
IPv4	IPv4	✓
IPv6	IPv4	✓
IPv4	IPv6	✗
IPv6	IPv6	✗

Strategy 1: Outside in

DUAL-STACK ARCHITECTURE - IPV6 TARGET GROUP

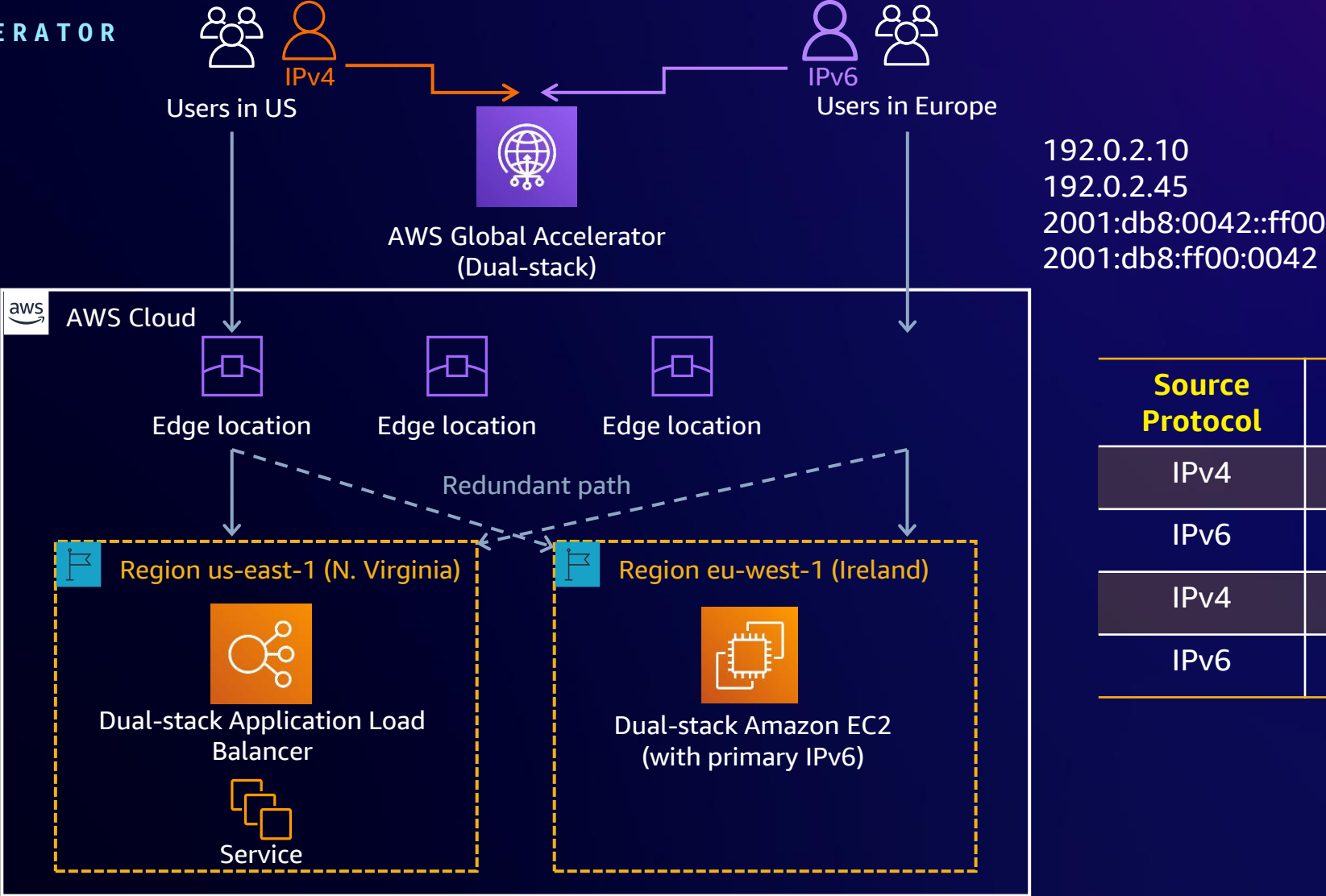


Source Protocol	Destination Protocol	
IPv4	IPv4	✗
IPv6	IPv4	✗
IPv4	IPv6	✓
IPv6	IPv6	✓



Strategy 1: Outside in

GLOBAL ACCELERATOR



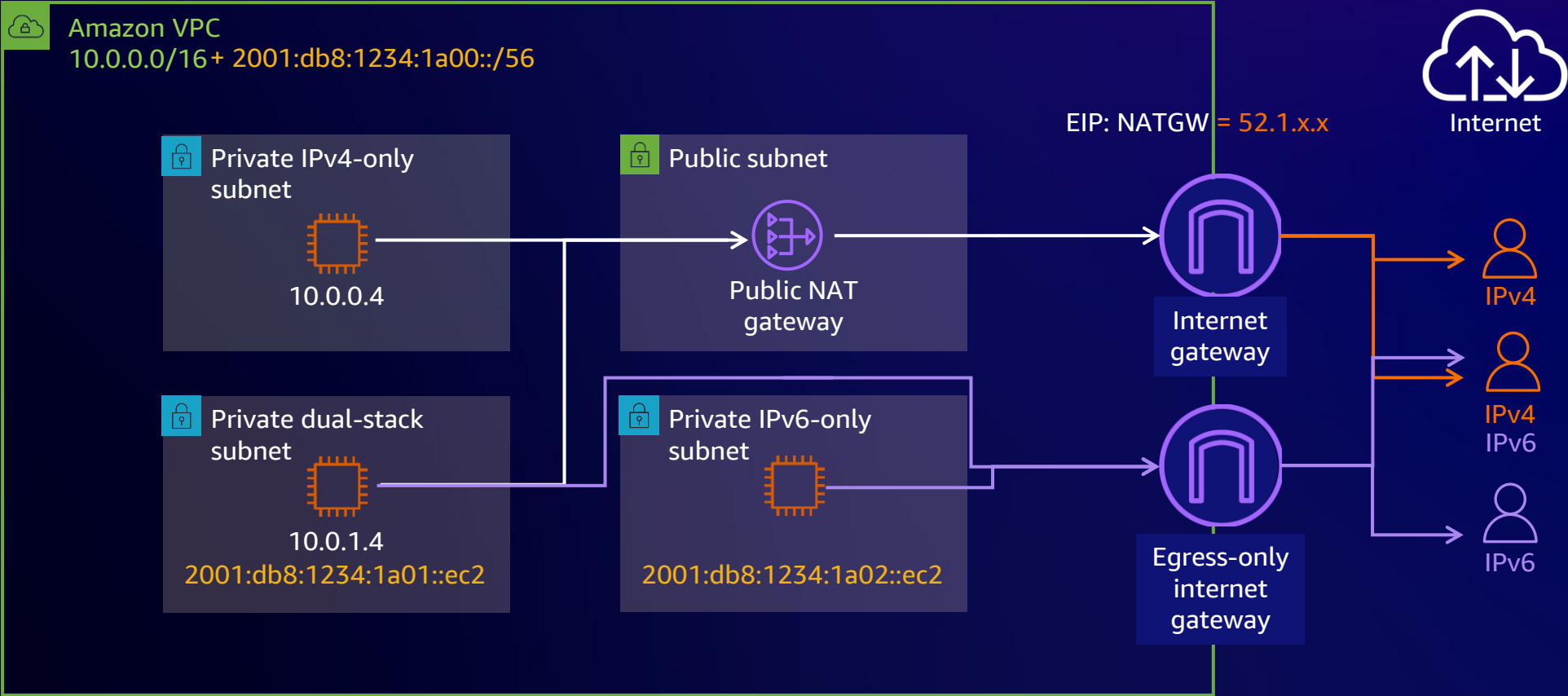
Source Protocol	Destination Protocol	
IPv4	IPv4	✓
IPv6	IPv6	✓
IPv4	IPv6	✗
IPv6	IPv4	✗

Strategy 2: Inside out

Source Protocol	Destination Protocol	
IPv4	IPv4	✓
IPv6	IPv6	✓
IPv6	IPv4	✓
IPv4	IPv6	✗

Destination	Target
10.0.0.0/16	Local
2001:db8:1234:1a00::/56	Local
0.0.0.0/0	IGW

Destination	Target
10.0.0.0/16	Local
2001:db8:1234:1a00::/56	Local
0.0.0.0/0	NATGW
::/0	EIGW



Strategy 2: Inside out



Source Protocol	Destination Protocol	
IPv4	IPv4	✓
IPv6	IPv6	✓
IPv4	IPv6	✓
IPv6	IPv4	✓

Thank you!



Please complete the session
survey in the mobile app

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